

Seungheon (Hoon) Yang

[hoonyang00@gmail.com] | [Irvine, CA 92617] | [(949) 419-7812] | [Github] | [Portfolio]

EDUCATION

University of California, Irvine

Bachelor of Science in Computer Engineering

Sep 2021 – Present

Cumulative GPA: 3.73/4.00

Honors & Memberships:

- IEEE-HKN Eta Kappa Nu Honor Society – Inductee
- Dean's Honor List (8 of 9 quarters)

Relevant Coursework: Computer Architecture, Engineering Data Structures and Algorithms, Advanced C Programming, Digital Systems and Logic, Computer Networks, Electronics I/II, System Software, Processor HW/SW Inter, Intro ML For ENGRS

PUBLICATIONS

- Sakib, S., **Yang, S.**, Al Faruque, M., and Jain, U. "Decoupling Trajectory Planning from Visual Grounding for Low-altitude Aerial Navigation." *IROS 2026* (Under review)

PROJECTS

Speed Violation Tracker | *Python, OpenCV, Ultralytics, Flask, JavaScript, HTML/CSS*

Mar 2026 – Present

- Developed an end-to-end CV pipeline for real-time vehicle detection, tracking, and speed violation capture from video and livestream using YOLO and OpenCV, improving display FPS by **3.4×** via decoupled inference and rendering into separate threads
- Estimated vehicle speeds via perspective transformation, mapping user-defined trapezoidal ROIs to real-world coordinates with manual distance calibration for accurate speed calculation
- Building a Flask-based web application with video upload, interactive browser-based ROI drawing, and Server-Sent Events(SSE) for real-time log streaming of detection results

ZotTARS | *Python, FastAPI, Ultralytics, Raspberry Pi*

Sep 2025 – Present

Software Subteam

- Architected a modular FastAPI backend by mapping ROS-style node-based decoupling to the STT–LLM–TTS pipeline, enabling independent services for a real-time voice AI chatbot system
- Building a low-latency real-time control environment integrating PS5, GUI, and Arduino into a unified system
- Integrating YOLO-based object detection into a robotic assistant, enabling vision-grounded LLM responses and control decisions

Autonomous Spider Bot (AI Project) | *PyTorch, Python, PyBullet, Linux*

Jan 2026 – Jun 2026

- Owned sensing and compute integration for a multi-legged spider robot platform, mounting and configuring LiDAR and IMU sensors on an NVIDIA Jetson Orin Nano with secure remote access via Tailscale
- Engineered Proximal Policy Optimization (PPO) Algorithm for autonomous robot navigation and validated the learned policy in PyBullet simulations

MyToDo | *Python, Flask, MongoDB, Jinja2, HTML/CSS*

Dec 2025 – Jan 2026

- Implemented a task management web application with full CRUD operations using Flask and MongoDB
- Designed server-rendered pages using Jinja2 templates to create, update, and delete tasks

Drone Project | *Embedded Systems, UAV Hardware, ArduPilot*

Sep 2025 – Dec 2025

- Assembled UAV hardware, including the flight controller, ESCs, motors, GPS, batteries, and radios, with soldering and validation
- Configured ArduPilot and ESP32 telemetry for GPS-guided autonomous waypoint missions with real-time ground-station communication

EXPERIENCE

Undergraduate Research – AICPS Lab

Sep 2025 – Present

Research Assistant (Supervisor: Sadman Sakib, Ph.D. Student)

Irvine, CA

- Built a VLN evaluation and visualization pipeline in OpenUAV for trajectory analysis, metric extraction, and video generation
- Operationalized and validated our proposed AeroLANCE navigation architecture on a physical UAV by conducting structured task-based evaluations across multiple navigation scenarios
- Reimplemented the TypeFly baseline within the same codebase to perform controlled comparative experiments
- Performed monocular 3D reconstruction from physical UAV image sequences using MonST3R and generated visualizations in Blender
- 3D-printing and assembling the CRAFT Hand, a tendon-driven robotic hand, using provided 3MF files and assembly documentation

SKILLS

Programming languages: Python, C/C++, Java, JavaScript, HTML/CSS, SQL (Oracle), Arduino C/C++, LC-3 Assembly

ML/CV & Robotics: PyTorch, OpenCV, ROS2, Ultralytics, Reinforcement Learning, Computer Vision, Autonomous Navigation

Tools & Libraries: Linux, Git, Flask, FastAPI, Jinja2, PyMongo

Languages: Native in Korean, Fluent in English